

P2: the zeros of B near roots of unity with $\ell \leq 0$ ($|1 - \zeta| \leq \frac{1}{2}$)

research programme P2, 2026-09-24. Not reviewed.

Notation of P1f_lemma.tex (Section “Setting and the saddle data”) and P1h_lemma.tex. $B(q) = \sum_k (-2(1-q))^k q^{k^2} / (q; q)_{2k}$, $\zeta = e(a/N)$, $\gcd(a, N) = 1$, $q = \zeta e^{-t}$, $t = re^{i\theta}$, $c = -2(1 - \zeta)$, $L = \text{Log } c = \ell + i\varphi$, $\ell = \log(4 \sin \pi \frac{a}{N})$, $\delta := |c|^{N/2} = e^{N\ell/2}$. The saddle data w , x_m , $V_m = \frac{L_m^2}{4} + E_w$, $G^*(m)$ are those of P1f Lemma saddle, but now for *both* roots w of $(1-w)^2 = c^N w$. Labels: PROVED; *computer-assisted* (finitely many Arb inequalities, script fails loudly); VERIFIED (numerics, not a proof); HEURISTIC; OPEN; CONDITIONAL.

Outcome.

- (A) **Why the zeros were not found.** The P1f modes have $\text{Re } x_m \approx \ell/2 < 0$ when $\ell < 0$: they leave the half-plane, as the row of RESEARCH_PROGRAM.OPEN.md says. The saddles that matter for $\ell \leq 0$ are different. They are a pair $\pm x_*$ exponentially close to the endpoint $x = 0$ ($k = 0$), with $|x_*| \approx \delta/(2N)$ and heights $\pm V_*$, $V_* = \sigma c^{N/2}/N^2 (1 + O(\delta^2))$, $\sigma = \pm 1$. So the zeros, if any, sit at $|t| \asymp |V_*| \approx e^{N\ell/2}/N^2$, which is exponentially small in $N|\ell|$ (for example $|t| \approx 6 \cdot 10^{-9}$ at $\frac{1}{25}$). This is PROVED for the pairing (Lemma 2) and VERIFIED for the size.
- (B) **The answer depends on the parity of N .**
- N odd: both members of the pair have $G^* \neq 0$, so $B \approx A_+ e^{V_*/t} + A_- e^{-V_*/t}$. The zeros accumulate at ζ along the tie ray $\text{Re}(V_* e^{-i\theta}) = 0$, which is $\theta^* \approx \pm \frac{\pi}{4}$, at $1/t_j = (\log(-A_-/A_+) + 2\pi i j)/(2V_*)$. This is VERIFIED at $\frac{1}{13}, \frac{1}{15}, \frac{1}{17}, \frac{2}{31}, \frac{1}{21}, \frac{3}{41}, \frac{1}{25}$: the zeros sit where predicted, to relative error 10^{-4} to 10^{-10} , and the argument-principle count matches the prediction (21 against 21.26). It is PROVED, computer-assisted, for individual zeros (Proposition 4): for example there is exactly one zero of B in an explicit disc of radius $3 \cdot 10^{-5}$ about $t = 3.5486 \cdot 10^{-4} + 3.6006 \cdot 10^{-4}i$ at $\zeta = e(\frac{1}{13})$. The accumulation itself (infinitely many zeros) is HEURISTIC with strong VERIFIED support. It is not proved.
 - N even: exactly one member of the pair has $G^* \neq 0$ (Corollary 3), so $B \approx H e^{V_*/t}$ is a single exponential. The zeros do not accumulate at ζ inside any sector $|\theta| \leq \frac{\pi}{2} - \eta$. This is HEURISTIC, and VERIFIED (argument principle in MPFR, $|\theta| \leq 1.2$): no zeros at $\frac{1}{20}$ ($10^{-6} \leq |t| \leq 10^{-2}$), $\frac{1}{16}$ ($10^{-5} \leq |t| \leq 10^{-2}$), $\frac{1}{12}$ ($2 \cdot 10^{-4} \leq |t| \leq 0.05$), $\frac{1}{14}$ ($10^{-4} \leq |t| \leq 0.05$), $\frac{1}{18}$ ($10^{-5} \leq |t| \leq 10^{-2}$) and $\frac{1}{22}$ ($10^{-6} \leq |t| \leq 10^{-3}$). On the real ray $|B|$ is constant to 10^{-5} , $\log |B| = 0.82318$ at $\frac{1}{16}$ for $10^{-6} \leq t \leq 10^{-3}$, because $V_* \in i\mathbb{R}$.
- (C) **Accumulation at every point of the small arc: CONDITIONAL.** The accumulation set is closed, and the odd- N points are dense. So if the odd- N mechanism holds on a dense set of $a/N \in (0, 0.0804)$, then every point of the arc $|\arg q| \leq 2\pi \cdot 0.0804$ is an accumulation point of zeros, even- N points included. At even- N points the approach is then tangential (through nearby odd- N points), never along a ray. What is missing is Theorem-M-type non-vanishing of the two amplitudes $A_{\pm} = \frac{1}{2}(H_e \pm H_o)$ (finite sums, Section 2) for a dense set of N , together with an error bound for the model. OPEN.

- (D) $q \rightarrow 1$. `paper2a` (Proposition `prop:infpolesalt` with Lemma `lem:polefinite`) proves that the zeros of B accumulate at $q = 1$ along $(0, 1)$, with $B \approx \cos \sqrt{2/t}$ (two exponentials $e^{\pm i\sqrt{2/t}}$). This is the degenerate member $N = 1$ ($c = 0$) of the odd case: two conjugate exponentials, with zeros on a tie ray. The exponent $\pm i\sqrt{2/t}$ replaces $\pm V_*/t$ because $c = 0$ (HEURISTIC link).

1 The landscape when $\ell \leq 0$

Lemma 1 (the two roots; PROVED). $c = 4 \sin(\pi \frac{a}{N}) e^{i(\frac{\pi}{2} + \pi \frac{a}{N})}$, so $c^N = |c|^N e^{i\pi(\frac{N}{2} + a)}$. Hence $c^N \in \pm i \mathbb{R}_{>0}$ for N odd, $c^N > 0$ for $N \equiv 2 \pmod{4}$, and $c^N < 0$ for $4 \mid N$. The roots $w, 1/w$ of $w^2 - (2 + c^N)w + 1 = 0$ are: off the unit circle and non-real for N odd; real positive, $w < 1 < 1/w$, for $N \equiv 2 \pmod{4}$; conjugate and on the unit circle for $4 \mid N$ and $|c|^N \leq 4$ (real negative for $4 \mid N$, $|c|^N > 4$, the P1f regime).

Proof. $1 - \zeta = -2i \sin(\pi \frac{a}{N}) e^{i\pi a/N}$. For $N \equiv 2 \pmod{4}$, a is odd and $\frac{N}{2} + a$ is even. The discriminant is $c^N(c^N + 4)$. If $|w| = 1$ then $w + 1/w = 2 + c^N$ is real, which is impossible for $c^N \in i\mathbb{R} \setminus \{0\}$. For $c^N = -\epsilon$, $0 < \epsilon \leq 4$, the discriminant $\epsilon^2 - 4\epsilon \leq 0$. $\square$

Lemma 2 (the landscape is odd; PROVED). Let $\Omega_m(x) = xL_m - x^2 - \text{Li}_2(e^{-2Nx})/N^2 + \pi^2/(6N^2)$ (P1f), and let $\tilde{\Omega}_m$ be its continuation to $\text{Re } x < 0$, with the principal Li_2 (cut $[1, \infty)$). For $\text{Re } x > 0$, $0 < |\text{Im } 2Nx| < \pi$ and $s = \text{sgn } \text{Im } x$,

$$\tilde{\Omega}_{m-s}(-x) = -\Omega_m(x).$$

Consequently, if x is a critical point of Ω_m (root $w = e^{-2Nx}$), then $-x$ is a critical point of $\tilde{\Omega}_{m-s}$ (root $1/w$), with value $-V_m$. The partner mode $m - s$ has the opposite parity to m . The same holds on $\text{Re } x = 0$ ($|w| = 1$) by continuity.

Proof. Put $u = 2Nx$, $z = e^{-u}$, $|z| < 1$. Inversion gives $\text{Li}_2(z) + \text{Li}_2(1/z) = -\frac{\pi^2}{6} - \frac{1}{2} \text{Log}^2(-z)$ for $z \notin (0, 1]$. Here $\text{Log}(-z) = -u + i\pi s$ when $0 < |\text{Im } u| < \pi$. Then $\Omega_m(x) + \tilde{\Omega}_{m'}(-x) = \frac{2\pi i(m-m')x}{N} - 2x^2 + \frac{\pi^2}{3N^2} + \frac{\pi^2}{6N^2} + \frac{(u-i\pi s)^2}{2N^2} = \frac{2\pi i x(m-m'-s)}{N}$, which vanishes for $m' = m - s$. Critical points and values follow by differentiation. $\square$

VERIFIED (`P2_landscape.py`, 40 digits): for all 14 cases of the table, $|x_* + x_{\text{partner}}| \leq 3 \cdot 10^{-41}$ and $|V_* + V_{\text{partner}}| \leq 6 \cdot 10^{-41}$, with the partner mode $m \pm 1$.

Corollary 3 (parity rule; PROVED from Lemma 2 and P1f Lemma saddle (iv)). The saddle values come in pairs $\pm V$. For N odd both members have $G^* \neq 0$. For N even exactly one member has $G^* \neq 0$ ($G^*(m) = 0$ unless m is even for $4 \mid N$, or odd for $N \equiv 2 \pmod{4}$).

Reality of V_ (VERIFIED, not proved).* $V_* \in i\mathbb{R}$ for $4 \mid N$, $|c|^N \leq 4$, and $V_* \in \mathbb{R}$ for $N \equiv 2 \pmod{4}$ (to 10^{-40} , `P2_landscape.log`). A proof would combine Lemma 2 with the conjugation symmetry that exists for even N : $\bar{L} = L - 2i\varphi$ and $\varphi N/\pi = \frac{N}{2} + a \in \mathbb{Z}$, so $\Omega_m(x) = \Omega_{\bar{m}}(\bar{x})$. With $-x_* = \bar{x}_*$ (case $|w| = 1$), one gets $-V_* = \bar{V}_*$ if the two partner modes coincide. That mode bookkeeping is not done here. For N odd, $\arg V_* \approx \arg c^{N/2} \equiv \pm \frac{\pi}{4} \pmod{\pi}$ (Lemma 1 and the size estimate below), so the tie ray is $\theta^* \approx \pm \frac{\pi}{4}$, well inside $|\theta| < \frac{\pi}{2}$.

Size (VERIFIED). Expanding $\text{Li}_2(e^{-u}) = \frac{\pi^2}{6} + u \log u - u + O(u^2)$ at the small root $w = 1 + O(\delta)$ gives $V_* = \sigma c^{N/2}/N^2 (1 + O(\delta^2))$ (HEURISTIC expansion). The measured relative deviation is about $0.014\delta^2$: $7.9 \cdot 10^{-3}$ at $\frac{1}{13}$ ($\delta = 0.75$) and $4.4 \cdot 10^{-10}$ at $\frac{1}{25}$ ($\delta = 1.8 \cdot 10^{-4}$).

2 The block model

HEURISTIC derivation. In the window $|t| \asymp |V_*|$, the saddle $x = kt \approx \delta/(2N)$ is tiny, so $p^j \approx 1$ except in the factors $1 - q^{Ni} = 1 - p^{Ni} \approx Nit$. Write $2k = Nn + r$, $0 \leq r < N$. With $\prod_{j=1}^{N-1} (1 - \zeta^j) = N$ this gives $b_k \approx c^k \zeta^{k^2} / (n! (N^2 t)^n (\zeta; \zeta)_r)$. The dropped factors cost $\exp O(N|t|n^2 + |x|^2/|t|) = \exp O(N\delta + \delta)$ at $n \asymp |V_*|/|t|$. Summing over n :

- N even: $B \approx H e^{\epsilon c^{N/2}/(N^2 t)}$, where $H = \sum_{\rho < N/2} c^\rho \zeta^{\rho^2} / (\zeta; \zeta)_{2\rho}$ and $\epsilon = 1$ for $4 \mid N$, $\epsilon = -1$ for $N \equiv 2 \pmod{4}$ (from $e(aNn^2/4)$).
- N odd: $B \approx H_e \cosh \frac{s}{N^2 t} + H_o \sinh \frac{s}{N^2 t}$, $s = c^{N/2}$, where $H_e = \sum_{\rho \leq \frac{N-1}{2}} c^\rho \zeta^{\rho^2} / (\zeta; \zeta)_{2\rho}$ and $H_o = s^{-1} \sum_{\kappa=\frac{N+1}{2}}^{N-1} c^\kappa \zeta^{\kappa^2} / (\zeta; \zeta)_{2\kappa-N}$. Even n give the cosh; for odd n , r is odd. So $A_\pm = \frac{1}{2}(H_e \pm H_o)$.

This is the “ $\cos \sqrt{2/t}$ ” mechanism of $q = 1$, transplanted to ζ : for N odd two exponentials; for N even one. VERIFIED (P2_heads.py against P2_zeros.py/P2_fitV.py; amplitudes are t -independent):

a/N	model $ A_+ , A_- $ (or $ H $)	measured	model V	measured V
1/13	1.4598, 1.4578	1.4671, 1.4766	$\pm(3.150 - 3.150i) \cdot 10^{-3}$	$(3.1737 - 3.1242i) \cdot 10^{-3}$
2/31	0.31936, 2.35318	0.32039, 2.35344		$(2.55993 - 2.55984i) \cdot 10^{-5}$
1/25	2.597302, 2.522243	2.597313, 2.522239		$(2.01922 - 2.01922i) \cdot 10^{-7}$
3/41	1.84041, 0.96455	1.84182, 0.95772		
1/16	2.30557	$e^{0.823183} = 2.27770$	$5.3719 \cdot 10^{-4}i$	$5.37328 \cdot 10^{-4}i$
1/20	2.84789	$e^{1.047612} = 2.85081$	$-2.30072 \cdot 10^{-5}i$	$-2.30081 \cdot 10^{-5}i$
1/24	3.42573	$e^{1.2312} = 3.4252$	$7.1231 \cdot 10^{-7}i$	$7.1111 \cdot 10^{-7}i$

The measured V agrees with the *exact* saddle V_* of P1f Lemma saddle to 10^{-9} relative. It agrees with the model value $\sigma c^{N/2}/N^2$ only to $O(\delta^2)$, as it should (P2_fitV.py, P2_fitV.log). The discrepancies of the amplitudes are $O(\delta)$ and $O(|t|)$ corrections.

3 Numerics (VERIFIED)

B is summed in MPFR (p2scan, Rust/rug), with the precision raised until 70 bits survive the cancellation $\max_k |b_k|/|B|$. That cancellation reaches e^{65} at $\frac{1}{13}$, $|t| = 6 \cdot 10^{-5}$. In plain f64 every run below $|t| \approx 10^{-5}$ is garbage.

- *Odd N , zeros on the tie ray* (P2_zeros.py): each predicted zero was refined by secant to $|f| \leq 3 \cdot 10^{-14}$. The relative deviation from the two-exponential prediction is $\leq 4 \cdot 10^{-4}$ ($\frac{1}{13}$ at $|t| = 5 \cdot 10^{-4}$), falling to 10^{-10} ($\frac{1}{25}$, $|t| \approx 2 \cdot 10^{-9}$). The arguments approach θ^* .
- *Count* (p2scan mcount, adaptive argument principle): $\frac{1}{13}$, $5 \cdot 10^{-5} \leq r \leq 2 \cdot 10^{-4}$, $0.6 \leq \theta \leq 1$: 21 zeros, against the predicted $(1/r_1 - 1/r_2)|V_*|/\pi = 21.26$.
- *Even N* : $\log |B| - \operatorname{Re}(V_*/t)$ is constant to 10^{-3} across $|\theta| \leq 0.6$ and $10^{-6} \leq r \leq 10^{-3}$ (P2_fitV.py). At $\frac{1}{16}$ this holds down to $|B| = e^{-302}$ at $\theta = -0.6$, $r = 10^{-6}$, so no second exponential is seen to that depth. The argument principle finds no zeros in any of the six even cases listed in (B) (P2_evenN_count.log).
- *Odd N , the switch* (P2_ray.py, $\frac{1}{13}$, $r = 10^{-4}$): $\log |B| - \operatorname{Re}(V_*/t) = 0.3832$ for $\theta \leq 0.7$. Past $\theta^* = 0.7933$, $\log |B| + \operatorname{Re}(V_*/t) = 0.39$, which is the partner $e^{-V_*/t}$ taking over.

4 Certified zeros (PROVED, computer-assisted)

Proposition 4. *Each disc listed in `P2_cert.log.txt` contains exactly one zero of $t \mapsto B(\zeta e^{-t})$ (winding number 1). In particular:*

- $\zeta = e(\frac{1}{13})$: $|t - (3.5485736701 + 3.6006300470i) \cdot 10^{-4}| \leq 3 \cdot 10^{-5}$ (48 arcs, $n_0 = 40$, 300 bits).
- $\zeta = e(\frac{1}{13})$: $|t - (4.3610782252 + 4.4297740295i) \cdot 10^{-5}| \leq 5 \cdot 10^{-7}$ (24 arcs, $n_0 = 40$, 300 bits, run in checkpointed chunks, `P2_cert_1_13.b.log`; worst $\text{rad}/|\text{mid}| = 0.423$, truncation index $K \leq 15322$, Cauchy majorant $\leq 3.9 \cdot 10^{32}$, which shows the cancellation, error $\leq 4.2 \cdot 10^{-5}$).
- $\zeta = e(\frac{1}{17})$: $|t - (2.5340248445 + 2.5392660205i) \cdot 10^{-5}| \leq 3 \cdot 10^{-6}$ (24 arcs, $n_0 = 40$, 300 bits, `P2_cert_1_17.log`). A first attempt at $|t| \approx 5.6 \cdot 10^{-6}$ with 16 arcs failed the $\text{rad} < \frac{1}{2}|\text{mid}|$ test (ratio 0.56) after $K = 146889$ terms per arc. It is kept as `P2_cert_1_17_failedM16.log`.

So there are zeros $q = \zeta e^{-t}$ with $|q - \zeta| < 6.3 \cdot 10^{-5}$ at $e(\frac{1}{13})$ and $|q - \zeta| < 3.9 \cdot 10^{-5}$ at $e(\frac{1}{17})$, exactly where the pair $\pm V_*$ predicts them.

Method (`P2_cert.py`). Naive ball evaluation over a t -ball fails, because the radius is multiplied by $\max |b_k|/|B|$ (up to e^{65}). Complex ball multiplication also inflates by $\sqrt{2}$ per product, so q^j and b_k are never formed by repeated multiplication. On each arc piece $|t - t_c| \leq \rho$ the script uses a Taylor model: $\log b_k = k(\log 2 + i\pi + \text{Log}(1 - q)) + k^2(2\pi i \frac{a}{N} - t) - \sum_{j \leq 2k} \text{Log}(1 - q^j)$ as an `acb.series` at the exact point t_c . Here $\text{Re}(1 - q^j) > 0$, so the principal Log has no cut, and $1 - q^j = -\text{expm1}(-jt)$ when $N \mid j$. The Taylor truncation is bounded by Cauchy on $|h| \leq R = 8\rho$ with the majorant $\sum_k \sup |b_k|$. That needs no cancellation, only an upper bound. The tail is bounded by the decreasing ratio bound $\rho_k = |c_t|Q^{2k+1}/((1 - Q^{2k+1})(1 - Q^{2k+2})) < \frac{1}{2}$. Each image enclosure must avoid 0 with $\text{rad} < \frac{1}{2}|\text{mid}|$. Two consecutive images then lie in a sector of opening $< \pi$, and the principal argument differences add up to $2\pi \cdot \text{winding}$. $\square$

These are finitely many zeros. They prove that zeros exist at exponentially small $|t|$ where the pair predicts them. They do not prove accumulation.

5 What would close P2

1. *Odd N , accumulation at ζ :* an asymptotic $B = A_+e^{V_*/t} + A_-e^{-V_*/t} + o(1)$ on the tie ray, uniformly as $t \rightarrow 0$, and $A_+A_- \neq 0$. The block model is accurate only for $N|t|n^2 \ll 1$, i.e. $|t| \gg N|V_*|^2$. Below that, the full saddle V_* (P1f Lemma saddle, both roots) takes over. The saddle is within $O(\delta/N)$ of the endpoint $x = 0$ and on the far side of $\text{Re } x = 0$ for one member ($|w| > 1$), so the P1f contour and the Lemma LS estimates do not apply as they stand. OPEN.
2. *A dense set:* non-vanishing of $A_{\pm} = \frac{1}{2}(H_e \pm H_o)$ uniformly on a dense set of odd N . These are finite sums with near-cancellation, the same kind of object as Z_N in Theorem M. With that and (1) at the level of the window $|t| \asymp |V_*|$ (where the model error is $O(N\delta)$, exponentially small), Rouché would give one zero near each such ζ . By closure that suffices for accumulation at *every* point of the small arc (item (C)). This is the shortest route: it needs no $t \rightarrow 0$ limit. OPEN.
3. *Even N , zero-free sector:* $B = He^{V_*/t}(1 + o(1))$ uniformly in $|\theta| \leq \frac{\pi}{2} - \eta$, and $H \neq 0$. OPEN.

Weakest points

- The mechanism is specific to $\ell \leq 0$ (more precisely to $\delta \lesssim 1$). At $\frac{1}{11}$ ($\ell = +0.12$, $\delta = 1.93$) the same fit fails: the fitted $|A_-| \approx 1.2 \cdot 10^5$ and the secant runs off to an unrelated zero. So other saddles (the P1f modes) matter there. The gap $0.0804 < \frac{a}{N} < 0.143$ is not addressed.
- Nothing about accumulation is proved. The certified statement is finitely many zeros per ζ .
- The model for even N and odd N is heuristic. Its agreement with the numerics is to $O(\delta)$ in the amplitudes and 10^{-9} in the exact V_* . There is no proof that the other $\pm$ pairs (for example $\pm 0.049i$ at $\frac{1}{16}$, which is admissible) contribute nothing. They are invisible numerically down to $|B| = e^{-302}$ and $e^{\pm 302}$.
- The zero-free claim for even N is sampled only on $|\theta| \leq 1.2$. The tangential directions, where zeros from nearby odd- N points must enter if (C) holds, are not sampled.
- The MPFR argument-principle counts use adaptive sampling, not a certificate.
- Lemma 2 is proved off the real x -axis; the $N \equiv 2 \pmod{4}$ case (x_* real) is by continuity from either side, where the two sides of the Li_2 cut differ. The numerical pairing check covers it.

Files. `p2scan/` (Rust: `f64` and MPFR evaluation, argument principle), `P2_B.py` (unused first Arb attempt, kept for the record of the $\sqrt{2}$ radius blow-up), `P2_fitV.py`, `P2_ray.py`, `P2_zeros.py`, `P2_heads.py`, `P2_landscape.py`, `P2_cert.py`, `P2_cert_chunks.sh`; logs `P2_fitV.log`, `P2_ray.log`, `P2_zeros.log`, `P2_heads.log`, `P2_cert_1_13.b.log`, `P2_cert_1_17.log`, `P2_log.txt`, `P2_landscape.log`, `P2_evenN_count.log`, `P2_certlog.txt`.